

Aditya Mehta

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EDUCATION

Master of Computer Science (GPA 4.00/4.00)

Arizona State University, Tempe, AZ

Aug 2023 – May 2025

Bachelor of Technology, Computer Science (GPA 7.78/10.00)

University of Petroleum and Energy Studies, Dehradun, IN

Aug 2016 – May 2020

PROFESSIONAL EXPERIENCE

Software Engineer Lead

Capgemini India — Client: General Motors (US)

Feb 2023 – Jul 2023

Bengaluru, IN

- Designed a serverless image pipeline with AWS Lambda, S3 triggers, and API Gateway, cutting turnaround by 40% and compute cost by 25%.
- Designed and deployed a Python-based parallel processing workflow on AWS (Lambda, SQS, Step Functions) that reduced batch runtimes by 60% and increased processing capacity fourfold with zero downtime.
- Paired Kubernetes autoscaler with AWS Spot Instances to absorb 4x traffic at 30% lower cost, backed by CloudWatch–Grafana predictive alerts that halved critical tickets.
- Migrated multi-repo workflows to a mono-repo CI/CD with environment-based Helm charts, enabling one-click dev-to-prod deploys.

Software Engineer

Larsen & Toubro Infotech — Client: Scania AB (Sweden)

Jun 2020 – Feb 2023

Bengaluru, IN

- Engineered a FastAPI service on PostgreSQL, delivered via Jenkins CI/CD; IsolationForest predicts traffic spikes and drives Grafana auto-scaling dashboards.
- Launched the SAPS API on API Gateway and AWS Lambda with DynamoDB and OAuth 2.0, empowering 600+ teams to self-serve and reducing support requests by 40%.
- Streamlined WSO2 upgrades - CodeBuild Docker images, CloudFormation IaC, Lambda-driven CodeDeploy, cutting release time by 70%.
- Built a PyTest pack for 120 REST endpoints with CloudWatch/SNS alerts, driving 99% deploy confidence.
- Synchronized CloudWatch logs to Datadog across three regions; scripted monitors and Jenkins alerts trimmed incident triage 30% while sustaining a 99.9% SLA.

Software Developer Intern

CaratLane India

May 2019 – Jul 2019

Chennai, IN

- Delivered an AWS-hosted task-tracker (Django REST, Angular) plus a nightly Python Lambda digest, unifying work queues for eight teams and saving 30 min per stand-up.
- Built a self-service portal listing Docker images in ECR, storing metadata in MongoDB; OAuth 2.0/JWT roles let non-developers manage containers, cutting DevOps hand-offs by 50%.

PROJECTS

Hybrid ML and Quantum Pipeline

MinIO, Redis Streams, OpenFaaS, Apache Airflow, TensorFlow, Qiskit, Docker

Feb 2025 – May 2025

- Benchmarks classical CNNs against quantum kernel SVMs on limited-sample image datasets.
- Architected a reproducible pipeline—**MinIO** storage, **Redis Streams** triggers, **OpenFaaS** functions, and **Apache Airflow** DAGs—containerised in Docker/k3s.
- Achieved 92% TensorFlow accuracy, executed **Qiskit/Aer** benchmarks 3× faster than prior scripts, and published results via **MLflow**.

FastAPI Microservice Suite on Kubernetes

FastAPI, PostgreSQL, Redis, Helm, Prometheus, Grafana, k3s

Sep 2024 – Dec 2024

- Project builds a scalable backend for high-freq IoT sensors that outgrew a Flask monolith.
- Deployed **FastAPI** services on single-node **k3s** with **Helm**; integrated **PostgreSQL**, **Redis**, and **Prometheus/Grafana** observability.
- Reduced p99 latency 40%, maintained under 150 ms at 4× load, and enabled one-click dev-to-prod rollouts via GitHub Actions.

MongoDB Analytics Dashboard

Feb 2024 – Apr 2024

MongoDB, React, Node.js, Docker Compose

- Delivers real-time makerspace tool-usage analytics to replace nightly CSV exports.
- Built a **React** dashboard fed by **MongoDB** change streams through a **Node/Express** WebSocket API; orchestrated with Docker Compose and deployed on Render.
- Live charts cut on-site wait times 25% and eliminated manual reporting for staff.

SKILLS

Programming & Scripting: Python, Java, SQL, Bash, TypeScript, React

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, MLflow, MLOps, Model Serving

Quantum Computing: Qiskit, Aer Simulator, Quantum Algorithms, Kernel Methods

Cloud & DevOps: Docker, Kubernetes, Terraform, GitHub Actions, MinIO, OpenFaaS, Apache Airflow

Architecture & Design: Microservices, REST APIs, Load Balancing, Distributed Systems, Agile Methodologies

Testing & Quality: PyTest, JUnit, Integration Testing, Test Automation

Databases: PostgreSQL, MySQL, MongoDB, CockroachDB, Redis